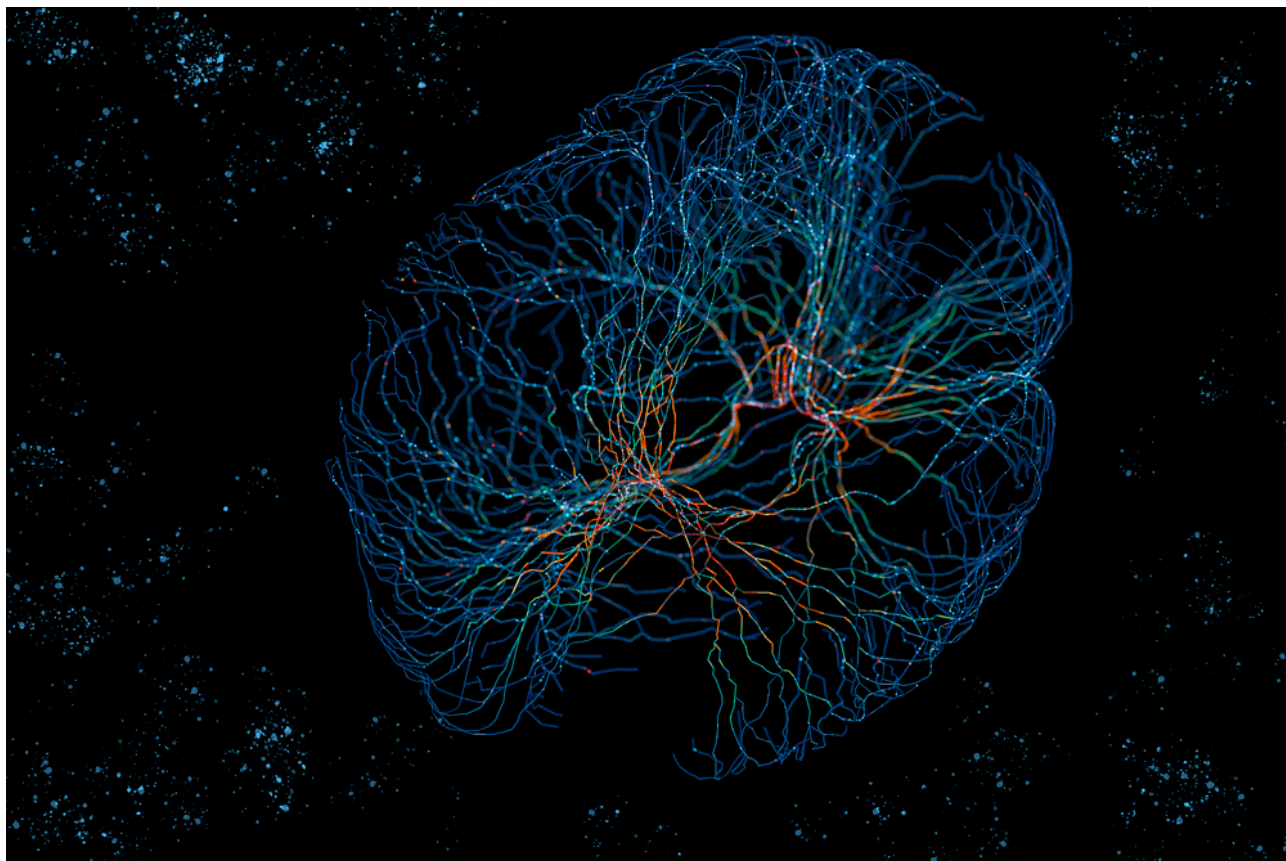


olinkWrapper

Preprocessing Exclusion Report



Report Date
February 26, 2026

Software Version
OlinkWrapper v1.0.0

App Developer
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Disclaimer: Please remember **olinkWrapper** is based on the **OlinkAnalyze** package from the Olink Developers. Please cite/use the original package.

 olink-wrapper.serve.scilifelab.se

 github.com/JD2112/olinkWrapper

Citation: Zenodo: 15098644

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1 Introduction

This report summarises all exclusion steps performed during the preprocessing of Olink NPX data.

2 Step 1: A. Data Preview – Exclude samples with QC Warning

Performed on: 2026-02-26 10:49:36

2.1 Background

Measurements flagged by Olink’s NPX Manager or NPX Explore with a QC Warning (anything other than ‘PASS’) are removed at the protein level to prevent unreliable data points from biasing the analysis.

2.2 Execution Details

Removed 1038 measurements flagged with QC warnings (QC_Warning != ‘PASS’). Proteins affected: BMP6, EPHX2, PGLYRP1, NPPB, TNNT3, HNRNPB, CEBPB, VIM, CHEK2, CRHR1, TSLP, MFAP3, TCL1B, DUOX2, CTF1, AKR1C4, HEBP1, SERPINB5, HK2, GRK5, VSTM2L, TIA1, CNPY2, CRX, VAMP5, SUSD1, RCOR1, TNF, CA5A, FABP6, STK11, TSPAN1, ACTA2, MTPN, EDIL3, ZBTB17, PTN, CEP43, IRAG2, COMT, MMP7, PON2, ADAMTS16, GP2, WASF1, MPHOSPH8, DNAJB8, SLITRK6, GPR37, GZMH, GYS1, THPO, CLEC1A, ENTPD6, IL6, CEACAM8, S100P, ENTPD5, ACE2, DCN, GLO1, PAG1, ADAM15, USP8, ITGB1BP2, CTSH, BAG6, MSTN, BOC, ADH4, CLTA, TP53INP1, CCL27, AK1, ANXA4, ACOX1, NT-proBNP, CLC, CNST, LEPR, TSHB, S100A11, VSIR, PCDH17, PDGFRA, CDH17, SNX9, NTRK2, ACY1, DOK2, AGXT, STK4, TGM2, DPP7, PLIN3, PLPBP, ICAM5, PRKAR1A, IL6ST, DCTPP1, PM20D1, LACTB2, FADD, CGREF1, CXCL8, LGALS1, IGFBP1, MND4, CDH6, IL19, ACAN, PLXNB3, TYRO3, EIF4EBP1, PDCD6, APLP1, CLEC5A, GSTA1, IGSF8, MEP1B, SDC1, CDH2, MCFD2, FCRL1, GPNMB, QDPR, GHRL, PPP1R2, CD2AP, NADK, CBLIF, TINAGL1, CANT1, SSC4D, OLR1, DIABLO, LRP11, SEMA3F, LEP, LPL, ADGRG2, SIGLEC7, NPTXR, GRAP2, CDHR5, THOP1, GLRX, KITLG, PILRB, CLUL1, AHCY, FABP2, REN, GDF2, RNASE3, SOST, NPDC1, CD69, CXCL5, COL4A1, LILRA5, MARCO, ANGPTL1, PRSS27, SORT1, SFTPD, DDC, SDC4, HMOX1, SNAP23, SERPINA12, GH1, SPON2, NECTIN2, CHRDL2, ROR1, CA13, CCDC80, FBP1, CTSL, NRCAM, HSPB1, THBD, KYAT1, FAM3C, TYMP, PCSK9, IGFBP7, UMOD, PLTP, MFAP5, LDLR, CA4, EPHB4, CDH5, CES1, FABP4, IGFBP1, PLAT, ICAM2, ADAMTS13, IL1RL1, GDF15, PLA2G1B, NOTCH3, LTBP2, TGFBR3, VWF, PEAR1, REG3A, XG, ADGRE5, FUCA1, CD209, MB, DKK3, TNFRSF10C, REG1B, IL2RA, PDGFRB, MET, ESAM, CA3, TNFSF13B, ALCAM, PLA2G2A, MSMB, CTSE, ICAM3, DPT, CRTAC1, CPA1, SERPINA11, CXCL16, SEMA7A, LGALS3, MEGF9, CORO1A, ENG, ENPP2, CPB1, SELE, PAM, COL6A3, CST6, CSTB, DLK1, GUSB, SCARF1, CELA3A, CCN3, HYOU1, LILRB2, ST6GAL1, TIE1, SIRPA, CASP3, PTPRS, CNTN1, BPIFB1, AZU1, PTPRF, NOTCH1, CHIT1, PDGFA, ACP5, ITGB2, FAS, PLXNB2, GAS6, EGFR, QPCT, RNASET2, SPP1, LILRB1, LILRB5, IGFBP2, PI3, RETN, CCL15, PRTN3, FAP, PPIB, COL18A1, AMY2A, CCL16, COL1A1, BLMH, PRCP, CD59, AMY2B, TCN2, SERPINE1, MCAM, DEFA1_DEFA1B, VASN, ART3, CNDP1, IL18BP, F9, ANPEP, SSC5D, ITIH3, CDH1, NCAM1, LBP, OSMR, HYAL1, CTSD, TNC, CD163, SPARCL1, NID1, AXL, PRSS2, CTSZ, CHL1, C1QTNF1, THBS4, TFRC, KIT, TFF3, F7, RARRES2, APOM, TIMD4, ITGB1, CD55, CD14, SELP, CD46, GP1BA, FCN2, HSPG2, PCOLCE, IL6R, SOD1, FCGR3B, TFPI, AOC3, NRP1, FCGR2A, LCN2, CR2, ADA2, CCL18, VCAM1, CD93, IGFBP3, CHI3L1, CST3, CCL14, PROC, ANG, FETUB, COMP, DPP4, ANGPTL3, PTGDS, CA1, C2, ICAM1, CCL5, IGFBP6, EFEMP1, REG1A, GGH, TGFBI, TIMP1

2.3 Samples Excluded (224)

1473924355, 1472550559, 1472532271, 2011599093, 2011538571, 1473871435, 2011619517, 2011563753, 1460114761, 2011585209, 1474003219, SC1_Run239, 2011605105, 2009861793, 1475104065, 2011471521, 2011490553, 2011486365, 1473905179, 2011517877, 2009814849, 2009846841, 1473579349, SC2_Run239,

2011600509, 1473873145, 1472135779, 1475104089, 2011594533, 2011583793, 1472137770, 2011527513, 2011484991, 2009843961, 1470704015, 2011614189, 2011633533, 1472546563, 2008993389, 1470670481, 2011490997, 1472116879, 2009782617, 2008980395, 2009858055, 1470724655, 2011567635, 2011492545, 1473873091, 2009151723, 1472501305, 2011503597, 1474009453, 1474001995, 2009805741, 2009858175, 1470669143, 2011502049, 2011576881, 2011569357, 1473872130, 2011471569, 2009862417, 1474010089, 1474011307, 2009847849, 2009821521, 1470704699, 1472568079, 2011593663, 2011492605, 1473871477, 2011601085, 1470747959, 1473578761, 2011495281, 1474002391, 1470759263, 1470702767, 1472505912, 2011502085, 2011532925, 2011564773, 1470706168, 1470754667, 2011550925, 2009785491, 1473579385, 1470667013, 1470713987, 1473959599, 1472136318, 2008993089, 2011499937, 2011594161, 1472133397, 1474060218, 1460107063, 1472132952, 1473874249, 1473872371, 1470740063, SC1_Run238, 1472135628, 2009151615, 2011627173, 2011473957, 1472133264, 1474060200, 2009863905, 2011477797, 1474044180, 1473901747, 1472560327, SC2_Run238, 1472135604, 2009153511, 2011615641, 2009776617, 1472090479, 1474015308, 2011503165, 1473890666, 1473873715, 1473903889, 1470736439, 1472136630, 2009153559, 2011571667, 2011521369, 2009000235, 1474028184, 1472571139, 1473887323, 1473875425, 1470717599, 2009792661, 2008995135, 2009030242, 2011571727, 2011516101, 1472132550, 2009000703, 2011468809, 1473993631, 1473887287, 1470717311, 2009791335, 1472134957, 2008986987, 2011617765, 2011623129, 1472548662, 1460112019, 1472084725, 1473875791, 1470714983, 2011611297, 1473655518, 2008987269, 2011625325, 1470715967, 1474059042, 2009030451, 2009159290, 1474012129, 1473872779, 1473887431, 2011638177, 2009116774, 2008998843, 2011639641, 2011524621, 1474059883, 1472132874, 2011598661, 1474057945, 1473872773, 1470715205, 2009869425, 1470721421, 2008999875, 1472119897, 1472073379, 1473578767, 2009118207, SC1_Run240, 1470734879, 2009156025, 1472118924, 1472069845, 2011576305, 1472135232, SC2_Run240, 1472107375, 1475103615, 1475117727, 1472119867, 1472071003, 2011549605, 2008993546, 2009156037, 1472109301, 1475104797, 1472069371, 1473584533, 2008993491, 1472136415, 1472119837, 1472070961, 2011575975, 1475102367, 2009156049, 2008999815, 1475102217, 1472068818, 1473584971, 2008993503, 2009154951, 1472118937, 1472099839, 2011640115, 1475102355, 2009150271

2.4 Proteins Excluded (369)

BMP6, EPHX2, PGLYRP1, NPPB, TNIN3, HNRNPK, CEBPB, VIM, CHEK2, CRHR1, TSLP, MFAP3, TCL1B, DUOX2, CTF1, AKR1C4, HEBP1, SERPINB5, HK2, GRK5, VSTM2L, TIA1, CNPY2, CRX, VAMP5, SUSU1, RCOR1, TNF, CA5A, FABP6, STK11, TSPAN1, ACTA2, MTPN, EDIL3, ZBTB17, PTN, CEP43, IRAG2, COMT, MMP7, PON2, ADAMTS16, GP2, WASF1, MPHOSPH8, DNAJB8, SLITRK6, GPR37, GZMH, GYS1, THPO, CLEC1A, ENTPD6, IL6, CEACAM8, S100P, ENTPD5, ACE2, DCN, GLO1, PAG1, ADAM15, USP8, ITGB1BP2, CTSH, BAG6, MSTN, BOC, ADH4, CLTA, TP53INP1, CCL27, AK1, ANXA4, ACOX1, NT-proBNP, CLC, CNST, LEPR, TSHB, S100A11, VSIR, PCDH17, PDGFRA, CDH17, SNX9, NTRK2, ACY1, DOK2, AGXT, STK4, TGM2, DPP7, PLIN3, PLPBP, ICAM5, PRKAR1A, IL6ST, DCTPP1, PM20D1, LACTB2, FADD, CGREF1, CXCL8, LGALS1, IGFBP1, MNDA, CDH6, IL19, ACAN, PLXNB3, TYRO3, EIF4EBP1, PDCD6, APLP1, CLEC5A, GSTA1, IGSF8, MEP1B, SDC1, CDH2, MCFD2, FCRL1, GPNMB, QDPR, GHRL, PPP1R2, CD2AP, NADK, CBLIF, TINAGL1, CANT1, SSC4D, OLR1, DIABLO, LRP11, SEMA3F, LEP, LPL, ADGRG2, SIGLEC7, NPTXR, GRAP2, CDHR5, THOP1, GLRX, KITLG, PILRB, CLUL1, AHCY, FABP2, REN, GDF2, RNASE3, SOST, NPDC1, CD69, CXCL5, COL4A1, LILRA5, MARCO, ANGPTL1, PRSS27, SORT1, SFTPD, DDC, SDC4, HMOX1, SNAP23, SERPINA12, GH1, SPON2, NECTIN2, CHRDL2, ROR1, CA13, CCDC80, FBP1, CTSL, NRCAM, HSPB1, THBD, KYAT1, FAM3C, TYMP, PCSK9, IGFBP7, UMOD, PLTP, MFAP5, LDLR, CA4, EPHB4, CDH5, CES1, FABP4, IGFBP1, PLAT, ICAM2, ADAMTS13, IL1RL1, GDF15, PLA2G1B, NOTCH3, LTBP2, TGFB3, VWF, PEAR1, REG3A, XG, ADGRE5, FUCA1, CD209, MB, DKK3, TNFRSF10C, REG1B, IL2RA, PDGFRB, MET, ESAM, CA3, TNFSF13B, ALCAM, PLA2G2A, MSMB, CTSB, ICAM3, DPT, CRTAC1, CPA1, SERPINA11, CXCL16, SEMA7A, LGALS3, MEGF9, CORO1A, ENG, ENPP2, CPB1, SELE, PAM, COL6A3, CST6, CSTB, DLK1, GUSB, SCARF1, CELA3A, CCN3, HYOU1, LILRB2, ST6GAL1, TIE1, SIRPA, CASP3, PTPRS, CNTN1, BPIFB1, AZU1, PTPRF, NOTCH1, CHIT1, PDGFA, ACP5, ITGB2, FAS, PLXNB2, GAS6, EGFR, QPCT, RNASET2, SPP1, LILRB1, LILRB5, IGFBP2, PI3, RETN, CCL15, PRTN3, FAP, PPIB, COL18A1, AMY2A, CCL16, COL1A1, BLMH, PRCP, CD59, AMY2B, TCN2, SERPINE1, MCAM, DEFA1, DEFA1B, VASN, ART3, CNDP1, IL18BP, F9, ANPEP, SSC5D, ITIH3, CDH1, NCAM1, LBP, OSMR, HYAL1, CTSD, TNC, CD163, SPARCL1, NID1, AXL, PRSS2, CTSZ, CHL1, C1QTNF1,

THBS4, TFRC, KIT, TFF3, F7, RARRES2, APOM, TIMD4, ITGB1, CD55, CD14, SELP, CD46, GP1BA, FCN2, HSPG2, PCOLCE, IL6R, SOD1, FCGR3B, TFPI, AOC3, NRP1, FCGR2A, LCN2, CR2, ADA2, CCL18, VCAM1, CD93, IGFBP3, CHI3L1, CST3, CCL14, PROC, ANG, FETUB, COMP, DPP4, ANGPTL3, PTGDS, CA1, C2, ICAM1, CCL5, IGFBP6, EFEMP1, REG1A, GGH, TGFBI, TIMP1

2.5 Pre/Post Filtering Summary

Metric	Before	After	Removed
Samples	224	223	1
Proteins	369	366	3
Measurements	82656	81618	1038

3 Step 2: B. Preprocessing > 3. LOD Detection – Filter proteins >50% below LOD

Performed on: 2026-02-26 10:49:46

3.1 Background

Assays where a significant portion of samples (typically >50%) fall below the Limit of Detection (LOD) are excluded. High proportions of missing or below-LOD data can reduce statistical power and lead to false discoveries.

3.2 Execution Details

Removed 13 protein(s) where >50% of measurements were below the Limit of Detection (LOD).

3.3 Proteins Excluded (13)

MTPN, HEBP1, ACOX1, TP53INP1, SERPINB5, DNAJB8, STK4, TCL1B, CTF1, CRX, DUOX2, CHEK2, HK2

3.4 Pre/Post Filtering Summary

Metric	Before	After	Removed
Samples	223	223	0
Proteins	366	353	13
Measurements	81618	78719	2899

3.5 Additional Notes

Threshold: >50% of sample measurements below individual protein LOD.

4 Step 3: B. Preprocessing > 4. Outlier Detection – Exclude outlier samples

Performed on: 2026-02-26 10:49:56

4.1 Background

Samples identified as multivariate outliers (using UMAP and distance-based heuristics) are excluded. These samples may represent technical failures or biological extremes that do not represent the common sample distribution.

4.2 Execution Details

Removed 12 sample(s) identified as outliers via UMAP-based distance analysis. Outlier threshold: mean + 2.5 x SD of pairwise distances.

4.3 Samples Excluded (12)

1474003219, 1473579349, 2009843961, 1470704015, 2009858055, 1470724655, 2009858175, 1470669143, 2009821521, 1470702767, 1470667013, 1470713987

4.4 Pre/Post Filtering Summary

Metric	Before	After	Removed
Samples	223	211	12
Proteins	353	353	0
Measurements	78719	74483	4236

4.5 Additional Notes

UMAP outlier threshold multiplier used: 2.5

5 Step 4: B. Preprocessing > 5. Manual Exclusion

Performed on: 2026-02-26 10:50:03

5.1 Background

User-defined exclusion of specific samples (e.g., control samples) or proteins based on study protocol or prior quality assessment.

5.2 Execution Details

Manually excluded samples and/or proteins. All control samples (SampleType CONTROL/CTRL/C) were excluded. Manual sample IDs entered: (none). Manual protein IDs entered: (none).

5.3 Samples Excluded (6)

SC1_Run239, SC2_Run239, SC1_Run238, SC2_Run238, SC1_Run240, SC2_Run240

5.4 Pre/Post Filtering Summary

Metric	Before	After	Removed
Samples	211	205	6
Proteins	353	353	0
Measurements	72365	72365	0

5.5 Additional Notes

‘Exclude all control samples’ checkbox was checked.

Report generated by olinkWrappeR on 2026-02-26 10:50:09.306774